

HONEYWELL SOFTWARE ENGINEERING TEAMS

AI Champions Programme

Module 10

Agent Prism for Monitoring, Observability and Enterprise Control

Going deep on traces, failure patterns, quality drift, and token/cost leak visibility — then closing the loop by connecting AI telemetry directly to implementation, review, and testing KPIs.

 DURATION 1.5 Hours	 FORMAT Hands-On Lab	 DELIVERED FOR Honeywell Eng. Teams
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How We'll Spend These Ninety Minutes

Six connected moves — from what a trace captures to telemetry tied to real KPIs.

01

Why Monitoring Agentic Engineering Matters

10 MIN

The gap between “the agent produced an answer” and “we know what it actually did.”

02

Observability Concepts: Traces, Replay & Failure Patterns

20 MIN

What a trace records, and why replay matters when a run goes wrong.

03

Token Usage & Cost Leak Visibility

15 MIN

The common, quiet ways agentic workflows burn tokens without adding value.

04

Quality & Control Signals: Spotting Drift

15 MIN

Telling a healthy agent from one that's quietly drifting off course.

05

Thresholds, Alerts & Operational Review

15 MIN

Why monitoring is a continuous cycle, not a one-time dashboard setup.

06

Hands-On Lab: Inspect, Compare, Connect to KPIs

15 MIN

Inspect real traces, compare runs, and tie findings to engineering metrics.

Module 10 in the 12-Module Journey

Agent Prism was introduced as a lens in Module 01 — this is where you actually learn to use it.

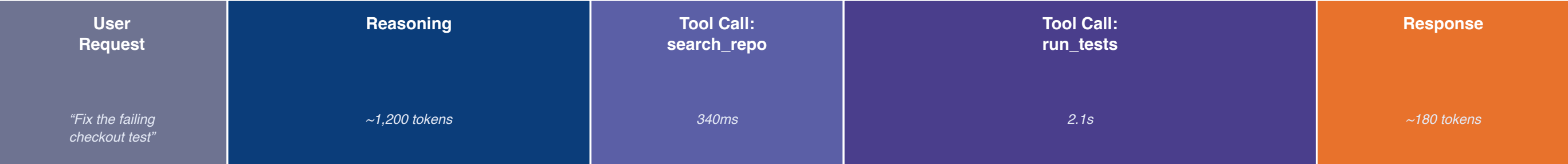


Why this matters: Module 11's ROI model is built directly from the cost and quality signals surfaced here; the capstone in Module 12 expects an Agent Prism monitoring design as part of its deliverable.

What a Trace Actually Captures

One agent run, broken into the steps Agent Prism actually records.

TRACE: fix-checkout-test-#4521



time →

Why this matters: Every trace is replayable — when a run goes wrong, you can step through exactly what the agent saw and did at each stage, not just the final output.

Quality & Control Signals: Spotting Drift

Drift rarely announces itself — it shows up as a slow trend across runs, not one bad answer.

HEALTHY

- Output style and structure stay consistent run over run
- Tool call patterns match what the task actually needs
- Confidence and quality scores stay within expected range
- Escalation rate to humans stays roughly stable

DRIFTING

- Output format quietly changes without anyone updating the prompt
- The agent starts calling tools it didn't need before
- Quality scores trend downward over consecutive runs
- Escalation rate climbs without an obvious cause

Token Usage & Cost Leak Visibility

Four quiet ways agentic workflows burn tokens without adding value.



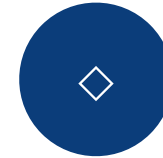
Redundant Context

The same repository or spec context resent on every turn instead of reused.



Retry Loops

A failed tool call retried repeatedly without backing off or escalating.



Over-Scoped Tools

A tool call pulling far more data than the task actually needs.

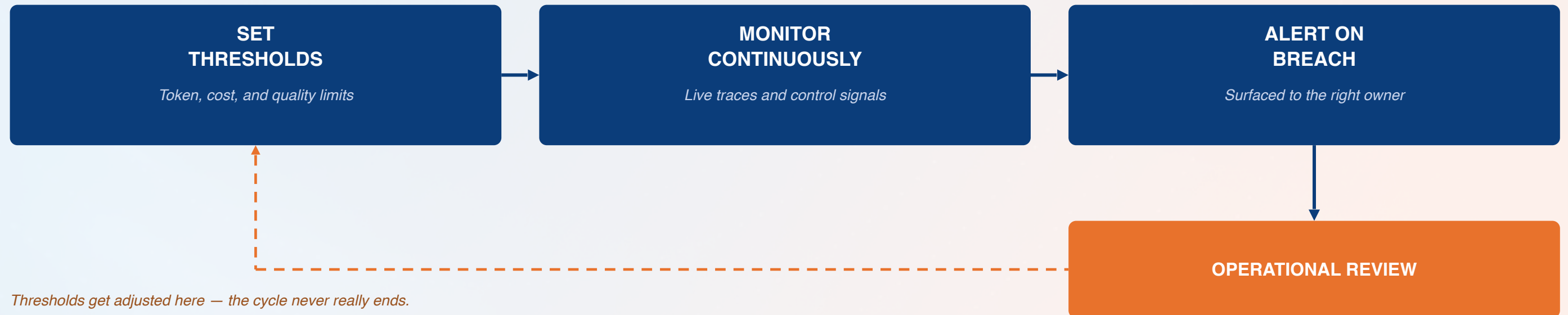


Unbounded Conversations

A session left open and accumulating context long after the task is done.

Thresholds, Alerts & Operational Review

A continuous cycle, not a dashboard you configure once and forget.



Connecting Telemetry to Engineering KPIs

Observability only matters once it changes what you measure.

Telemetry Signal	Connects To
Trace duration & tool-call count	Implementation cycle time (Module 01's baseline)
Failure & escalation rate	Quality / regression risk
Token usage & cost leakage	AI cost per feature (Module 11's ROI model)
Quality & control signal trends	PR review effort and confidence (Module 08)

Hands-On Lab Preview: Inspect, Compare, Connect to KPIs

The 15-minute activity that closes this module.



Ingest Run Data

Load or simulate engineering-agent run telemetry into Agent Prism.



Inspect Traces

Step through a run, span by span, to see what the agent actually did.



Identify Leakage

Spot token and cost leaks in the trace.



Compare Runs

Set a successful run against a poor-quality one, side by side.



Relate to KPIs

Connect what you found to implementation, review, and testing metrics.

The throughline: Observability only pays off once it changes a decision — this lab ends with you tying a trace to a KPI, not just looking at a dashboard.

Module 10 Outcomes & What's Next

- ✓ Why monitoring agentic engineering matters, and what a trace actually captures, is clear
- ✓ Healthy vs drifting quality and control signals can be told apart
- ✓ Common sources of token usage and cost leakage are identified
- ✓ The threshold → monitor → alert → review cycle is understood as continuous, not one-time
- ✓ AI telemetry has been connected to concrete implementation, review, and testing KPIs
- ✓ A real and a poor-quality run have been compared side by side in Agent Prism



NEXT MODULE

Module 11 — ROI, Token Economics and Deployment Governance (1 hour). Turning the cost and quality signals Agent Prism surfaces into a business case, and deciding when an AI workflow should be scaled, tuned, restricted, or retired.

Questions & Discussion

Module 10 — Agent Prism for Monitoring, Observability and Enterprise Control